

Timestamp 0 Hi, everyone, and welcome to week eight of the Human-AI Interaction course.

Today, we're going to enter a new kind of block of learning where we're going to talk about human-centered values and the design of AI systems.

And they're going to be quite a different set of values that we're going to discuss over this week.

Today, we're going to start first with automation and agency, and then we're going to look into fairness.

okay so the current rhetoric around ai often kind of has a singular focus on fully automated methods for example as a replacement of human work and human labor and to an extent this is kind of understandable right we are we we as humans are not really good at repetitive tasks we might have cognitive biases we might be have errors in judgment because of insufficient information We don't have big enough memories.

And we might even have ambiguous intents that

Timestamp 1 minute AI systems might be able to refine, right?

But an extreme focus in automation is also concerning for many reasons.

One of this was about to do with fairness, which we will discuss a little bit later. But another kind of important reason relates to the loss of critical domain expertise that

humans have and then
the AI system might not.
and we're going to talk about
two well-known examples of this
one known as google flu trends
and the other one about boeing
so originally human computer
interaction in human
computer interaction
there was this kind of
misleading one-dimensional
thinking suggesting that
designers have to choose
a place between human
control and computer
automation and would typically
consider that automation
was a better option.

So if we imagine this as a
timeline or as a line, then
you would have from one
side the human control and
the other side the human
automation and designers were
getting into this mindset,
but they would have to choose
a yellow point in between
those two for the design.

however this kind of kind
of more one-dimensional
thinking has brought also
some problems and i'll bring
two examples of that here
so google flu trends uh
is a you can see a graph
of that from 2015 i think
it was a web service that
was operated by google and
it provided estimates of
flu activity for more than
25 countries basically what
it did is it aggregated
search query so and then it
tried to make predictions
about flu activity so if
people wrote things like
what are the symptoms of
the flu then perhaps this
would be taken into a
consideration that this person

might have those symptoms
and then it would try to kind
of make predictions about
the flu activity in that
country where that person
is from and this was first
launched in 2008 to help
predict outbreaks of flu.
And Google Flu Trends
was in the end though
predicting more than
double the proportion of
doctor visits for
influenza-like sicknesses
based on the real data that
the estimates they had from
reports from the Centers
of Disease control and
probation prevention the cdc
so called cdc and the united
states so the cdc had real
reports from laboratories
across the united states
and it was the google trends
were actually predicting
double the amount of the
people that the those reports
were saying and this was
actually despite the fact
that the google flu trends was
built to help predict those
reports so they were over
predicting it over on two
sides on on almost on a
scale a double scale so why
what happened here what failed
Timestamp 4 minutes so partially this was
because they were trying to
automate this process of
of generating these reports
and knowing where to send more
um support which hospitals
or which regions to send
more support but because
of the the secretive like
the non-transparent non
-transparent algorithms that
Google has and because they
had started to include
features like autosuggest which

altered like honest honest
let's say more search patterns
that made it made the
original modeling much much
less effective and kind
of influenced what the the
report was so this attempt
for automation kind of failed
perhaps more more crucially
because they had fatalities,
there was the failure of
Boeing's MCAS system that
led to two accidents in 2018
and 2019, and they killed
together around 346 people
across the two flights.

And the problem here was
purely related to the fact
that it was Boeing's
automation failure and the
lack of agency of the
pilots to be able to engage
and kind of save solve
that issue so there was an
airplane an mcas flight
control software system
that was activating when
there was a wrong kind of
angle of attack giving
incorrect data so in no the
reality was that the
plane was kind of straight
the re the reading of
the sensor was thinking
that the nose the
nose was up so then
what it It was reacting
to, it was continuously
pushing the nose of
the airplane down.

And then the pilots
couldn't overcome that.

They couldn't kind of change
the angle of attack of the
angle of the airplane
because it was continuously
Timestamp 6 minutes pushing it down. And that
was two problems. First, that
it was automated and that
they couldn't take manual

control. And the second thing was that they had no training on this. they had not been informed about the existence of this system, which also meant that the pilots didn't know what the issue that they were facing was.

So here this speaks to a double problem, right? One is the problem of automating without giving human control, but the other one is also the need for transparency in these kinds of automated or semi-automated systems.

okay so maybe not because necessarily of these as examples but because of examples like these more recent iterations of the diagram that are the one-dimensional diagram i showed you before try to make it into two dimensions into a two-dimensional framework with the goal of achieving safe reliable and trustworthy interactions with ai systems where there's a high level of human control but also a high level of automation for instance a pacemaker or an airbag might require low user control because of the need for rapid action or the the fact that the the person might not have the expertise on what to do that on how to react but a bicycle or a piano or other creative activities should remain under under this framework it says that they should remain under human mastery if not for anything else just because people actually enjoy doing them and then this diagram mentions that there is a

perfect crux of control and automation and for instance making better camera photos because of the automated systems of exposure and focus but the photo does still is made by a human pressing the click or things like an elevator that can be informed or smarter in a way but the the exit of where the participant the user actually goes to remains on the control of the human but it's also the job of designers to prevent failures from excessive automation like the airplanes examples we we mentioned before as well as excessive human control so things like we won't discuss this a lot but things like managing your energy grid system is something that would doesn't is not valuable in terms of it's not something that humans can easily kind of grasp and kind of control but the examples we mentioned also show how it will be increasingly important to accurately locate who is responsible when agency is distributed in the system and control over an action is mediated over time and space. So in a parallel case in March 18, 2018, a self-driving Uber car struck and killed a pedestrian crossing her bike with her bike in the middle of an Arizona roadway. At the steering wheel of this autonomous vehicle was a safety driver, like a safety driver was sitting there. And her job was to monitor the

car's systems and take over in case of an emergency.

Now, the safety driver now faces criminal charges of manslaughter. So the devastating accident that happened and killed the person actually brought the question to the front that had only been hypothetical until now. So if a driverless car hits somebody who or what is it to blame so analyzing these kinds of situations Madeleine Elish a scholar came up with the concept of moral crumple zones that is mentioned here also in the slide that describe how responsibility for an action may be misattributed to a human actor who had limited control of the behavior of an automated or autonomous vehicle system system, not only a vehicle, a system in general.

So the parallel that she's making here is that the crumple zone in a car is decided to absorb the force of impact in a crash. And in this case, this crumple zone becomes a human that bears the brunt of the moral and other implications that the autonomous system might be kind of actually responsible for.

So why I'm bringing this example here is because these problems of human AI interaction of how does a human and a machine kind of interact and how does agency and automation distribute over this kind of framework we're saying actually intersects with very real

aspects of life like ethics and law studies, right?

It's not a hypothetical scenario we're describing anymore, but it's actually, it has real implications.

Okay, but maybe to bring it down a notch because not everything is high-stakes situations like self

-driving cars and airplane autopilot systems, similar

Timestamp 11 minutes discussions on agency and automation are faced by

thousands of people daily when they decide to use or they decide if and how to use generative AI systems.

And I will give you two recent examples of research that are exactly unpacking this thinking process.

The first is about a study called To Use or Not to

Use, Impatience and Over-Reliance when using generative AI productivity support tools. So what this study is trying to do

is trying to understand when do people kind of decide to automate tools? When

do they decide to use automation tools like

generative AI? And the research questions are actually

two. Given the goal of maximizing productivity, how good are people in making decisions on when to use generative AI?

And then how do people decide when to use or not to use generative AI tools?

And how did they examine? They achieved that by giving people a task to do and then varying this latency and the error rate of the

simulated generative AI support tool to see how users decide to make or fail optimal decisions when deciding to rely on that tool or not. And actually, it's pretty fun. That's why I'm bringing this example as well to the course. The tool that the people were asked to use was an image grid as presented here with colors. So participants were asked to fill in whatever they see a 4, they would put the gray, wherever they would see a 1, they would put a yellow, and so on or whatnot, to be able to make this image that you see on the right. So there were two tasks, there were two options. You could do a manual fill, so you could go step by step and kind of fill in the grid with the colors, or you could use assisted fill with a generative AI tool that kind of generated the first version of the image, and then you would go and correct it. And there were various conditions of latency, like how long the support tool would take to give the response, and how correct or not were Timestamp 13 minutes the answers that they were actually giving. And they tested that with 508 participants. So here is some example of the manual fill where participants would go step by step and they would create this duckling at the end of the process, whereas the assisted fill then the

first version of it was kind of there has kind of a high error rate as you can see because the the images are not necessarily that the boxes are not necessarily correctly aligned and then the participant would go and continue and correct any mistakes so what they did is that they asked people to do first the manual version then they would need to kind of get the kind of indication of how much latency or error rate they had and then they would have another image where they would need to kind of make a decision about whether they want to repeat that again and what whether they would like to do that with assisted fill or they would do that manually and the very interesting things they found is that when the assisted fill is either very high-performing in both latency and error rate, or very low-performing, so if we have very smart AI generative AI system or a very dumb generative AI system, then the people make optimal decisions, and they very clearly know when they should be using it. But the problem happens when the performance of AI becomes a bit more moderate. So especially in conditions when you have high-performing AI, but also in one condition but low-performing in the other condition, then it becomes more ambiguous so in cases where there's a smart but quite very slow AI or a dumb but

very fast AI so in this case people tend to over rely on it rather than take action by themselves and doing it manually assisted so it kind of gives us an indication of how people end up kind of over relying in some cases and some type of tasks on the generative AI tools

The next kind of research project that I think is very interesting that also has to do with the fact like how do people use generative AI and how does that relate to critical thinking is coming from a group in mostly Microsoft, which is called the impact of generative AI on critical thinking, self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers, which is a large title basically to say that they surveyed a bunch of people. that work with knowledge so people that mostly work with computer systems computer scientists or other kind of knowledge workers like this and they asked them to tell to report back to with a survey to what extent did they use kind of generative ai tools to support their work and do they feel like the cognitive effort and the confidence effects of that actually helped so we had they had two research questions when and how do knowledge workers perceive and the inaction of critical thinking and when and why do knowledge workers

perceive decreased or increased effort for critical thinking due to generative AI they had a quite kind of intricate research plan so they sent a survey and they asked people to tell their uses of three of three different usages of generative AI tools that could be any of the tools that people are comfortable with and then they asked them to talk about what factors task factors so what kind of task were they're doing how confident they were in that task and then assessed how critical thinking was involved in that task so as they say here perceived in action of critical thinking and perceived effort in critical thinking and then they also did a part of a survey where they talked about user factors like how much they trust generative ai their gender the age occupation and the tendency to reflect on work and they analyzed this both quantitatively and qualitatively they found that a higher confidence in generative ai is confident is associated with less critical thinking so the more confident people are in generative ai that they're kind of tended they They have the tendency to do less critical thinking, even though the same people don't think that there's a lot of effort to do critical thinking. So that's the interesting part. Like, even though they

find it kind of less

effort to do so, they
still do less of it.

And higher confidence is
associated with more critical
thinking, even though that
this is actually perceived
as more effort to do so.

So this is actually a
very interesting finding
in terms of how we design
systems that kind of support
people to do critical
thinking, right? To reflect
critically on the works that
they're doing, even when
they're perceiving that
this is kind of something
like secondary or if not
secondary, something too easy
to actually bother to do.

and the the suggestions
they come here is the design
strategy should focus on
balancing the these two
aspects so ai tools should
incorporate feedback mechanisms
that help users understand
the reliability of ai
outputs and when to trust
the ai when to apply the
critical skills and they can
also enhance the awareness
of opportunities of critical
thinking opportunities
telling them when to
can forgo for it for
things that are important
or secondary or where
they should actually
spend more effort to
it for things that are
actually more crucial.

And the reason I'm bringing
these two studies is besides
the fact that they have
very interesting findings,
it's because they also
demonstrate how HCI research
and methods like the user

testing and surveys and interviews actually get implemented in research in HCI. So I find that is an interesting kind of double example of both interesting findings that relate Timestamp 19 minutes to things that you can learn from, but also the methodological contributions that they're having in terms of how you can evaluate systems like this.

Hi everyone and welcome to the second part of Human Centered Values in the Design of AI Systems, the second video of week eight. Now this video we're going to talk about the concept of fairness and how that relates to AI systems.

So with the spread of AI in different types of applications, safety and fairness have become significant issue for researchers and engineers.

Machine learning and AI are used in courts to assess the probability of a defendant recommits a crime we're going to talk about that a bit more it is used in medical fields and childhood welfare systems and in autonomous vehicles so fairness is something that is very crucial and central into this so in the context of decision making uh so ai in supporting decision making fairness is the absence of any prejudice or favoritism towards an individual or a group based on

their inherent or acquired characteristics So basically, an unfair algorithm is one whose decisions are skewed towards a particular group of people. And COMPASS is one of the first examples that came out that is an exemplar of an unfair system, a discriminatory kind of system where AI has been used and machine learning has been used. So COMPASS is a tool used by courts, or at least used to be used by courts in the United States, to make pre-trial detention and release decisions basically when people kind of are arrested then this tool would help judges to decide whether the person the defendant should be staying in detention or should they be released and what's the possibility of them kind of recommitting a crime so the the software which is called compass which stands for correctional offender management profiling for alternative sanctions measures the risk of a person to recommit another crime and the judges used to use that to decide whether to keep them in prison or to let them go on bail for example the problem was that with a very kind of innovative investigation into algorithms one of the first one that came out in this type from propublica they found that there's have clear bias towards african-americans and compass was way more likely to have higher false positive rates for african-american

offenders than caucasian offenders enforcing predicting them that they are going to be higher risk of recommitting a crime uh in compared in comparison to their caucasian kind of other people so and then also when compared to non-human non-expert human judgment it was discovered not to be any better than a normal human so it also didn't seem to be able to have some kind of unique possibilities prediction possibilities that they were meant to have here you see for instance two people the one was classified as low risk and high risk and then explaining whether that the fact that the one on the on the right was subsequent offenses had none whereas the other one had subsequent offenses three three subsequent drug possession charges so it was actually clearly biased towards a subset of the population and that's not the only example there is out there it says it's one of the first ones that was kind of dominant but there are many of them that appear and i'm sure you know some of them as well.

Here is an example of facial recognition software in digital cameras that over predicts Asians as blinking as you can see here in the image. So this is another type of discrimination where because the training data has been used and has been collected by one sample of the population then

it obviously is skewed
towards that population.

The point is that we should
think responsibly and
recognize that the application
of these tools and the
Timestamp 4 minutes subsequent decisions actually
affect people's lives and
therefore considering
fairness constraints is a
crucial task when designing
and engineering these types
of tools so that's important
for you for human ai
interaction and that's important
for you as engineers as
well something similar was
founded by Joy Buolamwini
where she called at that
point she was she called
like algorithmic bias
as the coded gaze where
she's the founder of
the algorithmic justice
league and has done
amazing work since her I
think it was her master
thesis and later PhD to
kind of unpack the hidden
biases that our algorithms
kind of contain so I
suggest you go look at the
supplementary material for a
very nice presentation of her
work but i mean in summary
what happened was that she
was using computer vision
algorithms like the out-of
-the-box ones in order to be
able to to make her i think
the project was to make
a robot more um a social
robot like a human typical
human computer interaction
project with a social robot
trying to interact with
somebody but the camera and
the computer vision algorithm
was actually not recognizing
her because she's darker

skinned until she put a white mask so that kind of got her into the discussion and then the deep investigation what happens when algorithms are actually biased towards some sort of population against the other typically until that point which was probably around 2016 like the majority of um of research around bias and harms of algorithmic systems were about allocated farm harms like the one that we talked about in the beginning like the compass kind of system where it's about when a system does this has decision making power and then withhold certain groups from an opportunity or a resource and as Kate Crawford mentions in her keynote like a lot of research has been done towards how we can make decision-making more transparent or more or less biased towards some populations over others but there's also another type of representation of harm in information systems which is like representational so this occurs when systems reinforce the subordination of some groups are along some lines of identity or race like class or gender or something like this. So in this case, allocative harms, as I said, have received a lot of attention because their effects are immediate and it's about decision making. It's about getting a loan or not. It's about getting, being assessed by a judge or not in some way. And

they're easier to formalize
and study in computer
science and as well as in
economics, right? Because
it's about resource sharing.
But representational
harms have long term
effects and they're not easy
to formally characterize
and they're very much
dependent on the time and
space of when we're talking.
But as machine
learning has become
part of how we make
sense in the world,
so through technologies
like how we search, how
we translate, the voice
assistance, representational
harms will leave an
imprint on our culture,
as Kate Crawford mentions,
and will influence
identity formation and
stereotype perpetuation.
And I'll give some
examples of how that is.
Like, for example,
Joey Boulamini's
example is part of that.
For example, Boulamini's
example of shades of
gray is one part of those
representational harms.
but I wanted to bring your
attention to this kind of
more broader scoped view of
how bias kind of can perpetuate
within traditional deep
learning systems and then
we're going to talk about
them in terms of machine
learning but a lot of these
things that were discussed now
are evident also in the
generative AI technologies
the issue is that their
solutions for fairness in this
case and bias mitigation

might be kind of different. So we'll start with this and then hopefully we can generalize a little bit more for more generative AI models. So Hures and Kutag made a framework of sources of harm throughout the machine learning lifecycle and this is something that I think is very useful because it kind of helps us pinpoint where in our process perhaps we should be more aware of what we're doing and they can they divide the the machine learning life cycle in two parts the data generation which is a top part of this diagram you see and the model building and implementation which is the bottom part so i'm not going to talk through all of this model you have this as a as a core reading for this week but i'm going to bring out some examples and some key parts of of of of this diagram so the first example they're bringing and of bias that there's in the machine learning cycle is historical bias so historical bias is interesting because it represents that even if data is perfectly measured and samples and it might even represent the current situation of the world it can still inflict harm on a population because it involves representational harms like for instance as we talked about before reinforcing a stereotype to a particular group as an example that is very typical and it's also in yours and your suggested readings that we have word embeddings

where we have vector
represented where word embeddings
are vector representations
that encode semantic
meaning and obviously as you
obviously know that they
are widely used in NLP
application but research has
shown that because word
embeddings have learned from
large corpora of text like
Wikipedia or web pages or
Google News they reflect
the real-world biases about
women and ethnic minorities
or minorities so for instance
there are some occupations that
are traditionally more
gendered like nurse or engineer
so in these kind of models
they're highly associated
with words that represent
women or men respectively
so that's a bias that
exists in our culture more broadly
and that's something that this is
kind of translates
into the machine that still
kind of perpetuates these
kinds of stereotypes.

The second type of bias
that they talk about is
representation bias. And in
this case, we talk about when
the development of the sample
you're using for the data
set represents only some
part of the population and
subsequently fails to
generalize for the other, for
other parts of the population.
And that can happen in
many ways, like defining,
when defining the target
population it doesn't reflect
the whole of it when defining
the target population it
contains underrepresented
groups for instance like a

subset of adults between
18 and 30 of 18 and 60
might include pregnant woman
but that might be a very
small percentage of um of that
population which kind of
be is underrepresented when
we're talking about generalizing
that kind of subgroup
and then it's also about
sampling from the target
population when the sampling
method is limited or uneven
as an example which i think
is also part of your your
readings is image net which
is a large there's a very
very used image data set
consisting of 102 1.2 million
labeled images and it's
intended to be used widely
like to contain all natural
images however it doesn't
sample evenly right like
the majority of the images
are from north american or
northern or western europe
with only like one or two
percent of the images coming
from places like china or
india respectively so that
research has shown that the
when you when used these
models to classify images
concerning certain objects or
people the the specific
research i'm referring to
talks about what does a bride
look like what does a groom
look like what as a wedding
look like then when the
images come from underrepresented
countries then it
doesn't it doesn't get
recognized as such because it's
very much centered on a
western kind of imagery context
another part is a mother
kind of bias is like the
measurement bias when occurs

when choosing collecting or computing features and labels to use in a prediction problem so in this case is like the compass example where the models for these things for them the data for the models often include a proxy such as arrest in this case in the compass model to measure crime or to Timestamp 13 minutes measure an underlying notion of riskiness so we cannot directly measure riskiness so we use a proxy which is previous arrest in order to make that decision but because in this case for instance minority communities are more highly policed then they're more likely to be arrested therefore this proxy might misrepresent might have measurement bias in terms of how their underlying notion of riskiness so this is a kind of an important thing to notice about when you're using proxies in in data collection now the last thing i i will talk about and the rest i think you should kind of explore a little bit in the readings in the suggested readings is about learning bias and this case is learning bias occurs when modeling choices amplify performance disparities across different examples in the data and that's something we're going to also kind of refer to next week when we talk about sustainability of our models Timestamp 14 minutes and here i want to bring that some studies are showing how prioritizing compact models for instance with

methods such as pruning they can amplify the disparities between data with underrepresented attributes and this happens because given the more limited capacity the model learns to preserve information with the most frequent features which is not necessarily the ones that are the underrepresented attributes finally this is a has been quite grim so far like that there's a lot of places as you can see where we where models can be um bias and bring harm to the world but the point is not kind of to uh scare you away from doing this more of to inform you of the different kind of considerations you have to have in uh when when you're developing and designing ai systems and two ways that have been kind of emblematic in a way to solving these kinds of biases or at least addressing them to an extent would be to increase transparency where transparency in AI systems means that they should not operate as black boxes instead the argument is that users should have insight into the AI's reasoning process and we've talked a little bit about that in the class when we were talking about user participation in modeling and the second thing where we're talking about explainability which refers to an ability of an ai system to justify their outputs in the way that humans can understand which we

also brought some examples
of that in participation
in the participation
week where we showed how
models can actually
explain their decision
making in order to allow
for more fair decisions
or at least for more
transparent decisions
okay so this has been the
class on automation agency
Timestamp 16 minutes and fairness hopefully
you have a lot to think
about there's a lot of
additional readings in this it's
more theoretical it's
not very hands-on but the
point of this is to help
you kind of think how and
reflect on how the choices
you make while designing
and the choices of make
while developing kind of
reflects back to the population
that you're targeting.
Thank you.